

Fraud Detection System

1. Introduction

Fraudulent financial transactions have become a major concern for banks, credit card companies, and online payment platforms. Detecting fraud accurately and quickly helps reduce financial losses and improves customer trust. Machine learning techniques provide an efficient way to identify suspicious transactions by analyzing historical transaction data.

This project presents a Fraud Detection System that uses machine learning algorithms to classify transactions as either legitimate or fraudulent. The project includes data preprocessing, exploratory data analysis, handling class imbalance using SMOTE, model training, evaluation, and deployment using Streamlit.

2. Objective

The primary objective of this project is to build a machine learning model capable of detecting fraudulent transactions with high accuracy and reliability.

The project aims to:

- Detect fraudulent financial transactions.
- Handle imbalanced datasets using SMOTE.
- Train a machine learning classification model.
- Evaluate model performance using multiple metrics.
- Deploy the model through a Streamlit web application.

3. Dataset Description

The dataset contains transaction records with various attributes related to each financial transaction.

Features

Feature	Description
transaction_id	Unique transaction identifier
amount	Transaction amount

transaction_hour	Hour when transaction occurred
merchant_category	Category of merchant
foreign_transaction	Whether the transaction is foreign
location_mismatch	Indicates location mismatch
device_trust_score	Trust score of the device
velocity_last_24h	Number of transactions in the last 24 hours
cardholder_age	Age of the cardholder
is_fraud	Target variable (0 = Legitimate, 1 = Fraud)

4. Methodology

The project follows a standard machine learning workflow:

Step 1: Data Loading

The dataset is loaded using the Pandas library.

Step 2: Data Cleaning

- Checked for missing values.
- Removed duplicate records.
- Verified data types.

Step 3: Exploratory Data Analysis (EDA)

Several visualizations were created to understand the dataset, including:

- Fraud vs Legitimate transaction distribution
- Transaction amount distribution
- Correlation heatmap
- Feature importance visualization

Step 4: Data Preprocessing

Categorical variables were converted into numerical values using Label Encoding.

Step 5: Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

Step 6: Handling Class Imbalance

Since fraudulent transactions are much fewer than legitimate transactions, the SMOTE (Synthetic Minority Oversampling Technique) algorithm was applied to balance the dataset.

Step 7: Feature Scaling

The numerical features were standardized using StandardScaler to improve model performance.

Step 8: Model Training

The following machine learning algorithms were used:

- Random Forest Classifier
- Isolation Forest

Step 9: Model Evaluation

The model performance was evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix

5. Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy

- Matplotlib
- Seaborn
- Scikit-learn
- Imbalanced-learn (SMOTE)
- Joblib
- Streamlit

6. Results

The Random Forest model successfully classified transactions into legitimate and fraudulent categories.

The evaluation metrics demonstrated that the model performs effectively in detecting fraudulent transactions while maintaining good precision and recall. The use of SMOTE significantly improved the model's ability to identify minority-class fraud cases.

The trained model was saved using Joblib and integrated into a Streamlit web application for real-time predictions.

7. Conclusion

This project demonstrates the application of machine learning techniques for fraud detection in financial transactions. By applying preprocessing, handling class imbalance with SMOTE, and training a Random Forest classifier, the model achieved reliable prediction performance.

The project highlights the importance of machine learning in improving fraud detection systems and reducing financial risks. The developed Streamlit application allows users to enter transaction details and receive instant predictions, making the system practical and user-friendly.

8. Future Scope

Future improvements for this project include:

- Implementing XGBoost and LightGBM models.
- Real-time fraud detection using streaming data.
- Deploying the application on cloud platforms.
- Integrating the model with banking systems.
- Building REST APIs using Flask or FastAPI.
- Adding explainable AI techniques such as SHAP for model interpretation.

